

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Construction Technology) (CBCS) Degree Examinations

May - June 2024

22CCT223 – BUILDING SERVICES AND MAINTENANCE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Define the following terms with respect to Fire and Life Safety.				
	i) Travel distance ii) Floor Area Ratio (FAR) iii) Fire Resistance	10	L*1	1	1
	iv) Fire lift v) Automatic sprinkler system				
b)	Briefly discuss the general exit requirements in case of fire hazard in buildings.	10	L2	2	2
2. a)	Discuss the requirements of fire alarm systems? Explain any two types of fire alarm systems.	10	L1	1	1
b)	Explain the following relating to fire safety in buildings.	10	L1	1	1
	i) Fire classification ii) Fire triangle				
3.	Discuss the maintenance, use, placement and testing aspects of fire extinguishers.	20	L2	2	2
Unit – II					
4.	Explain the features of following with appropriate sketches.	20	L2	3	2
	i) Traction elevators ii) Hydraulic elevators				
5. a)	Discuss minimum five design considerations connected to hot water system installation in buildings.	10	L1	4	1
b)	Discuss the inspection and testing of hot water installations.	10	L1	4	1
6. a)	Briefly discuss five renewable energy resources.	10	L1	4	1
b)	Explain the methods of solid waste disposal.	10	L1	4	1
Unit – III					
7. a)	Highlight the factors which influence the decision to carryout building maintenance.	10	L1	5	1
b)	Suggest the preventive measures and maintenance for the following:				
	i) Floorings ii) Doors and windows iii) RCC slab	10	L2	5	2
	iv) Drainage systems v) Painting				
8. a)	Discuss any five types of building maintenance.	10	L2	5	1
b)	Explain the following maintenance aspects in buildings.	10	L1	5	1
	i) Joint maintenance in concrete ii) Termite control				

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
